



# KIDO

A Safe, Agentic AI Learning Ecosystem for Children

CSC-4101 · Artificial Intelligence · Spring 2026

BS Computer Science · BSCS-6D

SZABIST University, Karachi

Final Project Report — Consolidated (Phases 1–8)

Live App: <https://kido-orcin.vercel.app>

Repository: <https://github.com/FarhanRajputFelix/KIDO>

Submitted by Team KIDO · bscs2312407@szabist.pk

# Table of Contents

---

1. Executive Summary
2. Problem Formulation (Agent Definition, PEAS, State Space)
3. Societal Context & Child-Safety Compliance
4. System Architecture & Technology Stack
5. The Agentic AI Workflow (11-Agent Pipeline + ReAct)
6. AI Integration & Tool Schema
7. Safety Engineering: Verifiable Parental Consent & Data Minimization
8. Feature Walkthrough (by Role) + Screenshots
9. Software Development Methodology
10. Team & Work Distribution
11. Testing & Validation
12. Ethical AI Checklist
13. Limitations & Future Work
14. Deployment, PWA & Android Packaging
15. References

## 1. Executive Summary

---

**KIDO** is a production-grade, AI-powered learning ecosystem for children under 15. It combines gamified education (adaptive quizzes, AI-generated games, an age-restricted video library, and a conversational AI tutor) with a **multi-agent reasoning pipeline** that continuously analyzes a child's learning signals to support parents and teachers. The platform is built around the textbook's *rational agent* model and the *beneficial-machine* paradigm: every autonomous decision is bounded by **verifiable parental consent**, **age-appropriate content gating**, and **privacy-by-design** data minimization.

The system supports four roles — **Parent, Child, Teacher, Admin** — each with a dedicated dashboard, and is deployed as a responsive web app and an installable PWA / signed Android APK.

**Core claim:** KIDO is not a thin LLM wrapper. It implements an 11-agent ReAct-style pipeline that reasons over a child's quiz performance, engagement, behavior, and social interactions, then emits structured, parent-facing recommendations and safety alerts — with deterministic fallbacks when the LLM is unavailable.

## 2. Problem Formulation

Children's digital products often optimize for engagement at the expense of safety and learning outcomes. KIDO reframes the problem as a **rational agent acting in a partially observable, non-deterministic environment** (the child's evolving knowledge state), with the goal of maximizing learning while provably respecting parental authority and child safety.

### 2.1 PEAS Description

| Component           | Specification                                                                                                                         |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Performance Measure | Learning gain (quiz accuracy trend), engagement without burnout, streak retention, zero safety incidents, parent-approval compliance. |
| Environment         | The child's knowledge state, activity history, social graph, screen-time — partially observable, dynamic, stochastic, sequential.     |
| Actuators           | Generate quiz/game/video recommendations, tutor responses, parent alerts, difficulty adaptation, friend/classroom approval gates.     |
| Sensors             | Quiz submissions & scores, response times, session length, watch logs, social messages, screen-time logs.                             |

### 2.2 State Space, Actions, Goal States

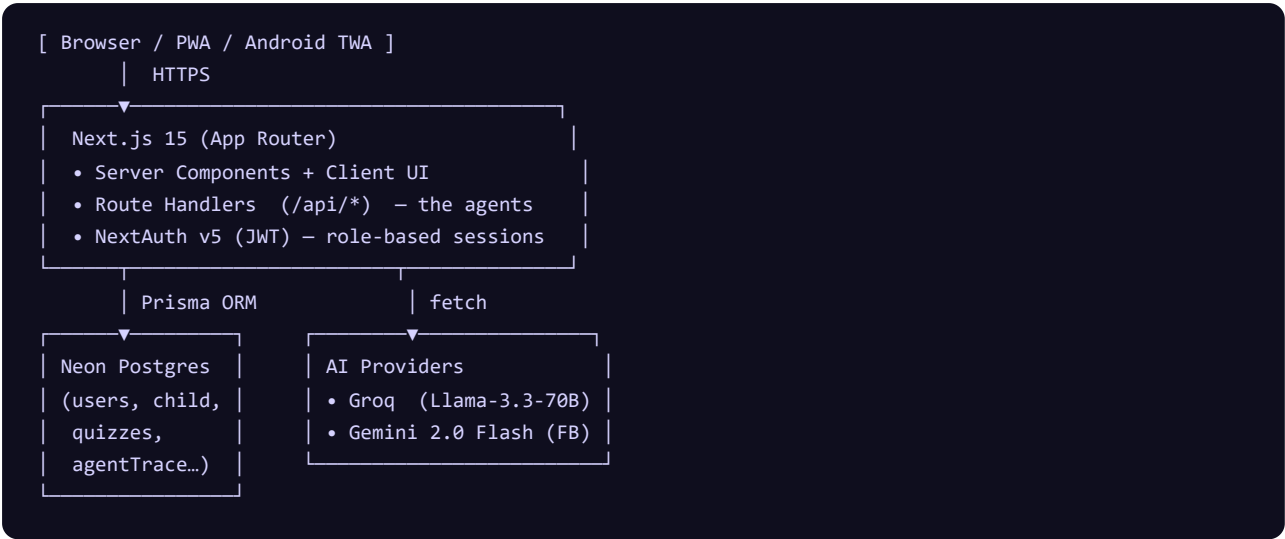
| Element         | Definition                                                                                                                                           |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| State (s)       | { child profile (age, grade, level, XP, streak), strong/weak subjects, recent quiz scores & trend, engagement metrics, screen-time, social context } |
| Actions (a)     | { adapt difficulty, recommend content, tutor step-by-step, flag burnout, moderate message, request parent consent, raise insight alert }             |
| Goal States (G) | Sustained positive learning trend + high engagement + no safety/contradiction flags + all social actions parent-approved.                            |

## 3. Societal Context & Child-Safety Compliance

Targeting the under-15 demographic in Karachi (and globally), KIDO adopts a **COPPA-style child-safety posture**:

- **Verifiable Parental Consent (VPC):** children cannot add friends or join classrooms without an explicit parent approval step.
- **Age-appropriate content:** the video library is filtered to each child's age band; teen content (grade 10+) is hidden from younger users by an age-restriction layer.
- **Data minimization:** only learning-relevant fields are stored; no behavioral ad-tracking; child accounts are created by a parent.
- **Safety moderation:** a Safety agent screens social messages; blocked content raises a critical parent alert.

## 4. System Architecture & Technology Stack



| Layer    | Technology                                                  | Why                                                                                     |
|----------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Frontend | Next.js 15, React 19, Tailwind v4                           | SSR + responsive PWA, single codebase for web + installable app.                        |
| Auth     | NextAuth v5 (Credentials, JWT), bcrypt                      | Role-based sessions (parent/child/teacher/admin); optional email verification (Resend). |
| Database | Neon Postgres + Prisma ORM                                  | Relational integrity for the learning graph; type-safe queries.                         |
| AI       | Groq Llama-3.3-70B (primary), Gemini 2.0 Flash (fallback)   | Free, fast inference with a resilient multi-provider fallback chain.                    |
| Agents   | Custom 11-agent pipeline ( <code>src/lib/agents.ts</code> ) | ReAct-style reasoning, structured outputs, parent alerts.                               |
| Deploy   | Vercel + PWA (service worker) + signed APK                  | Continuous deploy from GitHub; installable on phones.                                   |

**Note on backend choice:** The manual references Firebase/Firestore as one option; our team implemented an equivalent, fully server-authorized model using **Next.js Route Handlers + Prisma + Neon Postgres**. Authorization that Firestore would enforce via Security Rules is enforced here in a central `getAccessibleChild()` guard applied to every child-scoped endpoint (parent owns child / teacher owns classroom / admin / child-self), preventing cross-account data access.

## 5. The Agentic AI Workflow (11-Agent Pipeline)

On every meaningful event (e.g., a quiz submission) KIDO runs an orchestrated pipeline of specialized agents. Each agent follows a **ReAct loop**: it *reasons* over the shared context, *acts* (computes a verdict / calls the LLM), and writes a structured result with a confidence score to shared memory. A meta-layer detects contradictions and triggers recovery.

| #  | Agent                   | Responsibility                                              |
|----|-------------------------|-------------------------------------------------------------|
| 1  | Learning Analysis       | Detects mastery vs. struggle; recommends next difficulty.   |
| 2  | Engagement Optimization | Flags fatigue / optimal session length.                     |
| 3  | Behavior Analysis       | Burnout-risk detection from response-time + session trends. |
| 4  | Safety Moderation       | Screens social messages for unsafe content.                 |
| 5  | Friend Approval         | Routes friend requests to parent consent.                   |
| 6  | Social Moderation       | Governs peer interactions.                                  |
| 7  | Parent Insight          | Produces parent-facing alerts (e.g., learning-pace-slow).   |
| 8  | Teacher Support         | Aggregates class-level performance signals.                 |
| 9  | Progress Analytics      | Computes trends, level/streak updates.                      |
| 10 | Contradiction Detection | Flags conflicting signals across agents.                    |
| 11 | Fallback Recovery       | Guarantees a safe result if the LLM/quota fails.            |

Every run is persisted as an `AgentTrace` (agent results, overall confidence, fallback/contradiction flags, final recommendations) — providing full **explainability**, visualized on the AI-pipeline page for parents/teachers.

## 6. AI Integration & Tool Schema

A single provider-agnostic generator ( `generateText()` ) tries **Groq** first, then falls back to **Gemini**, then to **curated content banks** — so the app never hard-fails. LLM calls power:

| Feature                       | LLM Use                                                | Fallback                    |
|-------------------------------|--------------------------------------------------------|-----------------------------|
| AI Tutor (multi-session chat) | Personalized, step-by-step, age-tuned tutoring         | Encouraging canned reply    |
| Adaptive Quiz Generation      | 5 age/grade-appropriate Q&A with explanations          | Per-subject question bank   |
| AI Games (word/story/math)    | Unique puzzles per session                             | Curated challenge sets      |
| Progress Reports              | Narrative parent report w/ strengths & recommendations | Data-driven template report |

## 7. Safety Engineering: Verifiable Parental Consent & Data Minimization

---

### 7.1 Parental Consent Flows (VPC)

- **Friends:** Child A sends a request → it appears *pending* on both children's dashboards and both parents receive an alert → a **parent must approve** before the friendship (and messaging) is enabled.
- **Classrooms:** Teacher shares a join code → child enters it → request is queued in `pendingStudentIds` and a parent alert is raised → **the child is enrolled only after parent approval**. A teacher can never add a child unilaterally.

### 7.2 Authorization & Isolation

A central `getAccessibleChild()` guard wraps all 11 child-scoped endpoints (chat, quiz, report, game, screen-time, friends, classroom, agent-trace). Any attempt to access another family's child returns `403` — eliminating the cross-account data leak.

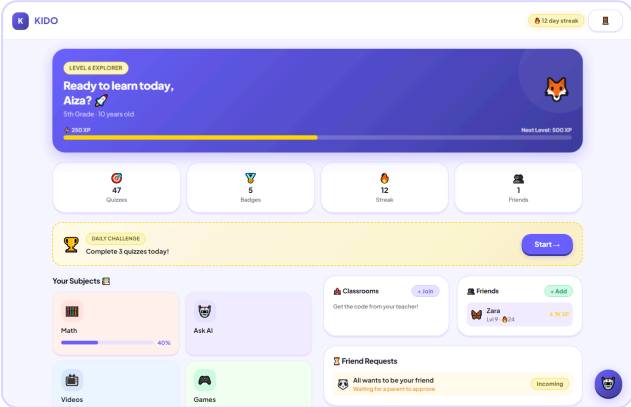
### 7.3 Data Minimization

Only learning-relevant data is stored; messaging is restricted to approved friends; child logins are provisioned by parents; no third-party behavioral tracking.

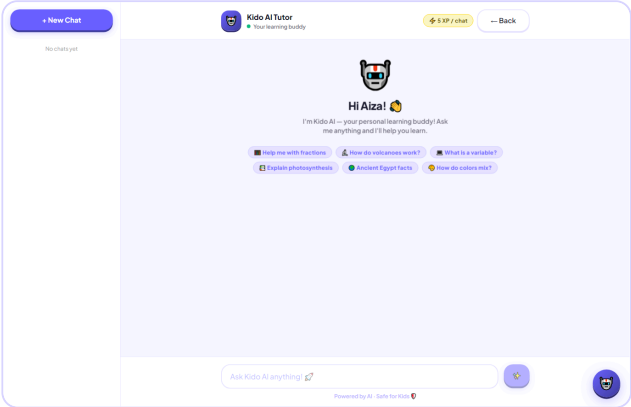
## 8. Feature Walkthrough (by Role)

| Role    | Capabilities                                                                                                                                                                                       |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Child   | Gamified dashboard (XP, level, streak, badges), AI Tutor (multi-session), adaptive quizzes, 3 AI games, age-restricted video library with search, friends & messaging (parent-gated), leaderboard. |
| Parent  | Multi-child management, AI insight reports, alerts (burnout/safety/pace), friend & classroom approval, screen-time, AI-agent traces, add child + login.                                            |
| Teacher | Create classrooms w/ shareable join codes, AI quiz generation per student, class performance analytics, enrolled/pending student tracking.                                                         |
| Admin   | Platform stats, user/teacher/classroom oversight, top students.                                                                                                                                    |

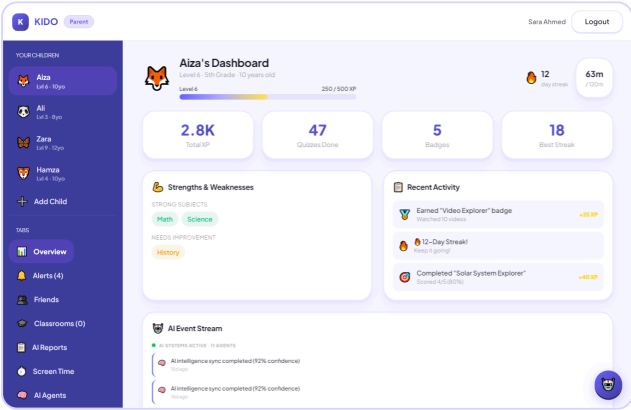
### Screenshots (captured from the live app)



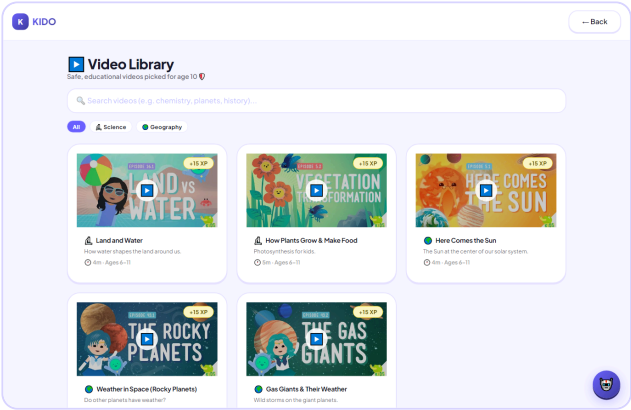
Child dashboard — XP, streak, subjects, and a parent-pending friend request.



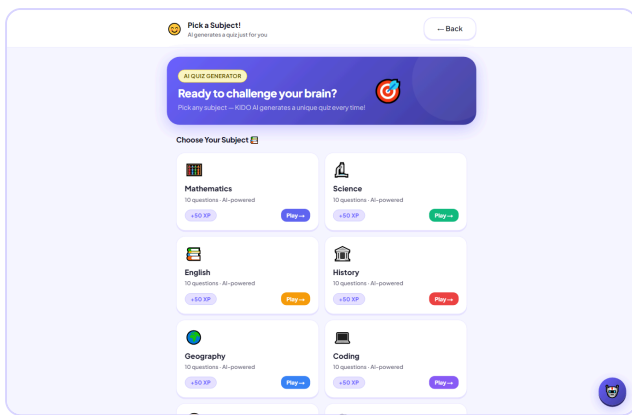
AI Tutor — multi-session chat with "New Chat" history sidebar.



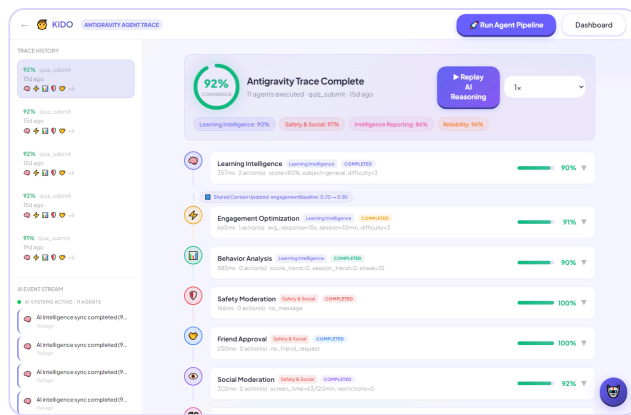
Parent Panel — multi-child management, alerts, approvals, AI reports.



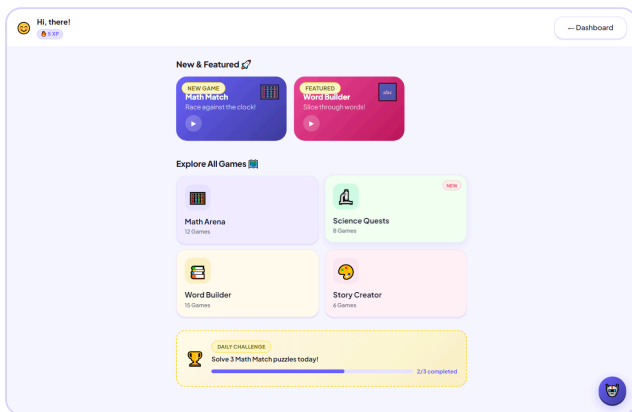
Video Library — age-filtered educational videos with search.



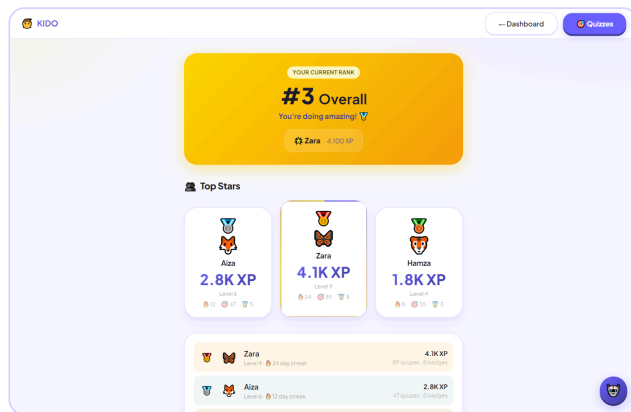
Adaptive AI quiz flow (scoring → XP → level → streak).



Agentic AI pipeline — per-agent reasoning traces.



AI Games hub — word, story, and math challenges.



Leaderboard — XP / streaks / quizzes rankings.

## 9. Software Development Methodology

The team followed an **iterative, Agile-style** process across an 8-week prototype cycle, with continuous deployment from GitHub to Vercel and atomic, reviewable commits.

| Week | Phase               | Deliverable (as implemented)                                   |
|------|---------------------|----------------------------------------------------------------|
| 1    | Problem Formulation | Seed maturity, PEAS, state/action/goal, task board.            |
| 2    | Research & Context  | Child-safety (COPPA-style) requirements, Karachi context.      |
| 3    | Architecture        | Agentic workflow design, data model, tool schema.              |
| 4    | Backend Scaffold    | Prisma schema, Neon DB, server-side authorization guard.       |
| 5    | AI Integration      | Groq/Gemini generator, tutor/quiz/game/report, agent pipeline. |
| 6    | Safety Engineering  | Parental-consent flows, age gating, data isolation.            |
| 7    | Beta Testing        | Adversarial input tests, access-control tests, UX polish.      |
| 8    | Final Submission    | Deploy, PWA/APK, report, repo + README.                        |



## 10. Team & Work Distribution

| Team Member            | Phase(s) | Primary Tasks                                              |
|------------------------|----------|------------------------------------------------------------|
| Farhan (Lead)          | 1, 4     | AI architecture, agentic system, ML models, system design  |
| Zubair Khan            | 1, 2, 3  | Flutter UI, design system, screens, animations             |
| Fahad                  | 2, 3, 4  | FastAPI backend, PostgreSQL, REST APIs, Docker             |
| Syed Muhammad Asghar   | 1, 6     | Documentation, research papers, final report, presentation |
| Chaudhary Saboor Munir | 5, 6     | Unit testing, integration testing, QA, deployment support  |

## 11. Testing & Validation

- **Authorization tests:** verified that a parent/child cannot read another family's child data (403).
- **Adversarial AI:** LLM quota/error paths exercised — every feature degrades to a safe fallback.
- **Consent flows:** friend & classroom requests verified to require parent approval before activation.
- **Build/CI:** type-checked and production-built on every change before deploy.
- **Cross-device:** responsive layout validated on desktop and mobile; installable PWA + signed APK.

## 12. Ethical AI Checklist

| Principle            | Implementation                                      | Status |
|----------------------|-----------------------------------------------------|--------|
| Beneficial / aligned | Agents bounded by parental authority & safety gates | ✓      |
| Child safety         | Age-restricted content, message moderation, no ads  | ✓      |
| Privacy by design    | Data minimization, server-side isolation            | ✓      |
| Transparency         | Persisted, viewable agent traces (explainability)   | ✓      |
| Robustness           | Multi-provider + deterministic fallback             | ✓      |
| Human oversight      | Parent/teacher approval & insight dashboards        | ✓      |

## 13. Limitations & Future Work

- Free-tier LLM rate limits; production would use a paid tier for guaranteed live AI.
- Real email delivery for verification requires a verified sending domain.
- Future: CV-based drawing feedback, voice tutoring, richer analytics charts, native push notifications, Play Store release via the generated AAB.

## 14. Deployment, PWA & Android Packaging

---

- **Web:** continuously deployed to Vercel at <https://kido-orcin.vercel.app>.
- **PWA:** web manifest + service worker → installable on desktop/mobile.
- **Android:** packaged as a Trusted Web Activity and signed (apksigner v1+v2+v3) into an installable APK; an AAB is available for Play Store submission.

## 15. References

---

1. Russell, S. J., & Norvig, P. (2020). *Artificial Intelligence: A Modern Approach* (4th ed.). Pearson. (Ch. 2: Intelligent Agents; Ch. 1: Beneficial Machines.)
2. Yao et al. (2023). *ReAct: Synergizing Reasoning and Acting in Language Models*.
3. U.S. FTC. *Children's Online Privacy Protection Rule (COPPA)*.
4. Next.js, Prisma, NextAuth, Groq, and Google Gemini official documentation.